The Media Connectivity Association




OVERVIEW AND UPDATES · NON-TERRESTRIAL NETWORKS (NTN)

# Non-Terrestrial Networks in 5G Systems

Expanding network connectivity via satellite and HAPS for content delivery

3GPP

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# About the project

The documents the project rests on

## 3GPP TR 38.811

**Technical Report**

The foundational NTN study item: the study on NR access to non-terrestrial networks, completed in Release 15.

## 3GPP TR 38.821

**Technical Report**

Solutions for NR to support non-terrestrial networks, the outcome of the Release 16 study on solutions for NR for NTN.

## 3GPP TS 38.300

**Technical Specification**

The NR overall description. Its NTN aspects define the feeder and service links, the payload types and the cell types.

FIND OUT MORE **Technology** **Specifications**

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# What NTN changes

NTN reuses the 5G system and the NR protocol stack; the differences are in the radio and in a few architecture roles

## System model where the NTN payload sits

Service link | Feeder link | NTN Gateway | Transparent payload | Regenerative payload | GNSS-capable UE

## Cells three types of service link

Earth-fixed | Quasi-Earth-fixed | Earth-moving

## Radio long delay and large Doppler shift

Timing advance | Common timing advance | Doppler pre-compensation | k-offset | Adapted HARQ, RRC and MAC timers

## System information assistance for the UE

SIB19: satellite assistance | SIB27: intended service areas of MBS broadcast

The NTN payload is on board a satellite or a high altitude platform station. SIB19 comes from the Release 17 NTN work and SIB27 from Release 19.

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# How it grew

The 3GPP work on non-terrestrial networks, release by release

| Rel. 15 | Rel. 16 | Rel. 17 | Rel. 18 | Rel. 19 | Rel. 20 |
|---|---|---|---|---|---|
| A study on NR for non-terrestrial networks (TR 38.811). | Studies TR 38.821 and TR 22.822. Only stage 1 of 5GSAT was defined. | First normative NTN: satellite RAT types, NR over transparent payloads with SIB19, NB-IoT and eMTC. | Satellite backhaul, discontinuous coverage, FR2-NTN bands and NTN-TN mobility. | Broadcast with an intended service area under one cell (SIB27), gNB on board, Store and Forward. | Release 20 (work plan of 28 September 2026), unverified: 5GSAT_Ph4, IoT_NTN_Ph4; TR 38.863 adds 3 MHz for n250 to n253. |

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# Mobility and MBS over NTN

What Release 19 specifies, as analysed on the Technology pages

## Mobility in a single NTN operator

- Time- and location-based measurement initiation
- Conditional handover: A4, T1, D1, D2
- Satellite switch with re-synchronization
- Feeder link switchover

## Deployment models the analysis's own grouping

- Single NTN operator
- Common NTN and TN operator
- Independent NTN and TN operators

## MBS broadcast the Intended Service Area

- 5G Core to NG-RAN
- SIB27
- MCCH
- UE behaviour

## MBS multicast NG-RAN decides

- PTP and PTM switching
- Multicast DRX HARQ timer for NTN
- Mobility of receivers

MBS multicast has no NTN-specific part in TS 23.247. For independent operators, MBS roaming is not supported in Release 19; Release 20 supports services of the visited network only.

WHERE WE STAND

# Non-Terrestrial Networks in 5G Systems

| BROUGHT TO STUDY | ON-GOING | OUTCOMES |
|---|---|---|
| OCUDU NTN Release 17 | | Study on handover and mobility over NTN |
| Uplift to Release 19 | | MBS over NTN profiling: broadcast, multicast and multicast mobility |
| MBS over NTN | | NTN overview: system model, radio-layer adaptations and system information |

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